ENVIRONMENTAL DATA ANALYSIS · CASE STUDY

# The cleanest city had the worst day

*What six US metro areas taught me about averages, wildfire smoke and the people who breathe the air (EPA monitoring data, 2020-2026)*

Sheila Alonso · October 2026 · Data: EPA AQS Data Mart, US Census Bureau (ACS 2022), HHS valuation parameters

Seattle has the cleanest air of the six metro areas I analysed: a mean Air Quality Index (AQI) of 53.5, the lowest of the group. It is also where I found the worst single day in the whole dataset. On 19 October 2022 the AQI reached 362, a level the EPA labels "Hazardous", because of fine particles from wildfire smoke.

> **An average tells you what a city is usually like. It says very little about the days that matter.**
>
> The idea behind this project

That contradiction is why I built this analysis. I wanted to know whether the cities with the worst average air are also the ones with the worst days, and how many people were actually breathing that air. The code and the check of every figure are in the notebook `air_quality_usa.ipynb`. This report tells the story for a non-technical reader, and the interactive Streamlit dashboard lets you explore the city comparison, trends and smoke events live.

| 6 | 515,685 | 362 | $58.1B |
|---|---|---|---|
| metro areas, each with a different pollution mechanism | monitor-level daily pollutant readings, 2020 to July 2026 | highest AQI recorded, in Seattle, the cleanest city on average | screening-level health cost across the six areas |

## Six cities, six different problems

I chose New York, Los Angeles, Houston, Denver, Chicago and Seattle because each is dominated by something different: traffic and heating, petrochemical industry, summer ozone, wildfire smoke. That makes the comparison about causes as much as levels. I used the EPA's daily composite AQI, which takes the worst pollutant on each day instead of an average, together with daily readings of PM2.5, PM10 and ozone, surface wind data, and county population from the Census (ACS 2022, summed over the counties included for each metro area, 22 of them for New York). The AQI series starts in January 2020 and ends between 28 February and 1 June 2026 depending on the city, and the pollutant readings run to late July 2026 (except New York's particulates, which end in spring).

Exposure is measured in person-days: the number of days above the EPA 24-hour PM2.5 limit (35 micrograms per cubic metre) multiplied by the population of each county. Trends are tested with the Mann-Kendall test, the standard choice for environmental time series.

## One city in a tier of its own

The Los Angeles area (labelled "California" in the project, since it is the only California region I studied) has a mean AQI of 88.8, against 53.5 to 69.1 for the other five. It had 247 days

above AQI 150 ("Unhealthy"), which is 10.7% of the period. Houston had 59 (2.6%), Denver 33 (1.4%), Chicago 28 (1.2%), Seattle 26 (1.2%) and New York 18 (0.8%).

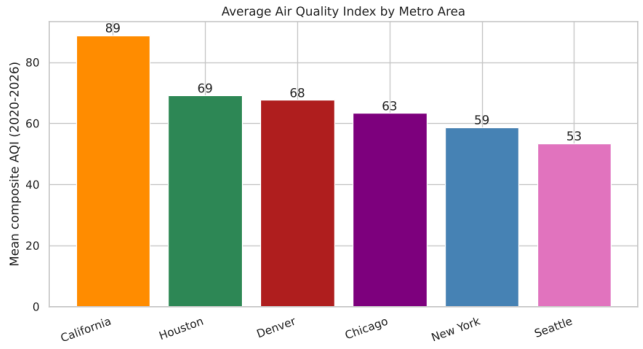



**Figure 1.** Mean daily AQI by metro area, 2020-2026

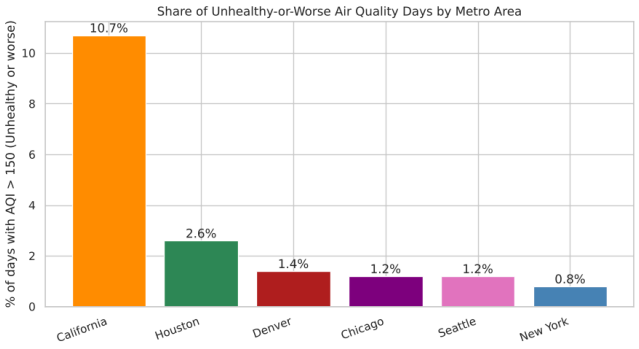



**Figure 2.** Share of days with AQI above 150

Its profile is a persistently high summer baseline in every year. Eight of its ten worst days fall in 2020: six are driven by ozone and two by fine particles. Its single worst day, 11 October 2021 (AQI 281), is defined by coarse particles (PM10), which points more to a dust event than to smoke, although the data records only the defining pollutant and not the cause.

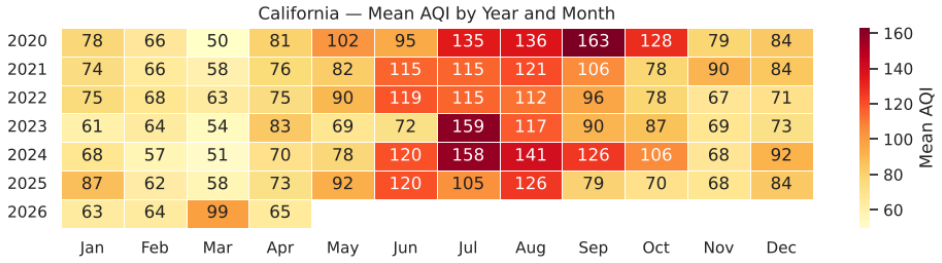



**Figure 3.** Los Angeles area: mean AQI by year and month

## Worst day versus worst pattern

A city can have clean air most of the time and still suffer extreme days, and the two stories need different responses. Seattle is the clearest example. Its worst days, in September 2020 and October 2022, are all driven by fine particles from wildfire smoke, and no ozone or coarse-particle day appears in its top ten. Los Angeles is the opposite: a high baseline all summer, every year. One is a city that is clean most of the year and is hit by short, extreme smoke events. The other is a city where the problem never really goes away.

## The smoke that crossed a continent

Seattle's 362 is only one side of the smoke story. In June 2023, Canadian wildfire smoke pushed New York to AQI 278 on 7 June and Chicago to 246 on 28 June, more than 1,000 kilometres apart. Surface wind data for those days does not point towards Canada. That fits what is documented about the event: the smoke travelled at altitude and settled locally, which ground-level stations cannot capture.

**WHY IT MATTERS**

A cross-border event with no local source produced values comparable to California's worst wildfire days. Local monitoring alone does not anticipate this kind of exposure.

## Counting people, not days

A bad day in Denver (about 2.9 million residents in the area studied) is not the same as one in Los Angeles (about 13.1 million). Once days are weighted by population, California dominates,

with 395.8 million person-days in 2020. But Seattle's 2020 smoke season produced 47.7 million person-days for only about 4 million residents, and the June 2023 smoke event produced 53.5 million in New York (6 to 8 June) and 17.0 million in Chicago (27 and 28 June), in just two or three days per city.

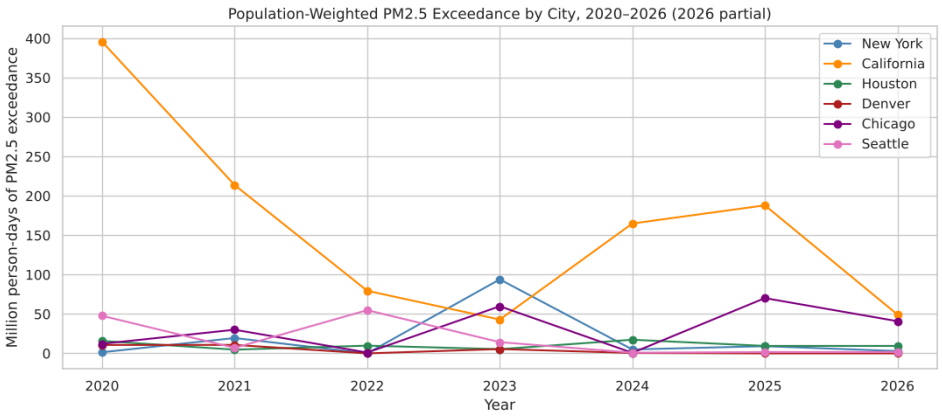



**Figure 4.** Population-weighted PM2.5 exceedance by city (person-days)

## Is the air getting better?

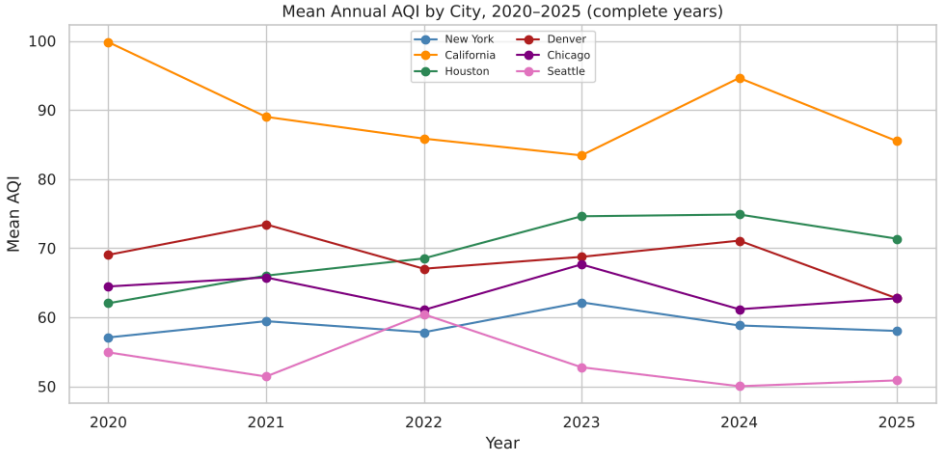



**Figure 5.** Mean annual AQI by city, complete years 2020–2025

The picture is mixed. California's mean AQI falls from 99.8 in 2020 to 85.5 in 2025, and Houston is the only city with a near-significant upward trend, in line with its higher values in 2023 and 2024. But no city has a statistically significant trend at the 95% level. Houston is the closest (Mann-Kendall $p = 0.056$, an upward trend), and California and Seattle trend downwards without reaching significance ($p = 0.272$). With six annual values the test has little power, so "not significant" means "not enough data to confirm", not "no trend". I left 2026 out of the test: it is only partly observed, without the summer and autumn months when the worst days happen in Seattle and the Los Angeles area.

If traffic drives pollution, air quality should be worse on weekdays. Four cities show a clearly higher AQI on weekdays: Houston (+4.5 points), Chicago (+4.3), Denver (+3.8) and New York (+3.4). California and Seattle show no difference, and they are the two cities whose worst days come from smoke, which has no weekly cycle. It suggests a traffic signal, but with six cities it does not prove the mechanism assigned to each one.

## What the exposure could cost

I combined the person-days with three published values: a 6% increase in mortality risk per 10 micrograms per cubic metre of PM2.5 (Pope et al. 2002), the US value of a statistical life

(USD 13.1 million, HHS 2024) and the national baseline mortality rate. The result is a screening-level estimate of about **4,438 attributable deaths and USD 58.1 billion** across the six areas in 2020-2026.

| Metro area | Estimated cost 2020-2026 | Attributable deaths | Cost per resident |
|---|---|---|---|
| California (Los Angeles area) | $27.3B | 2,081 | $2,079 |
| Chicago | $11.3B | 863 | $1,294 |
| Seattle | $9.8B | 750 | $2,455 |
| New York | $7.9B | 603 | $394 |
| Houston | $1.3B | 96 | $206 |
| Denver | $0.6B | 45 | $204 |

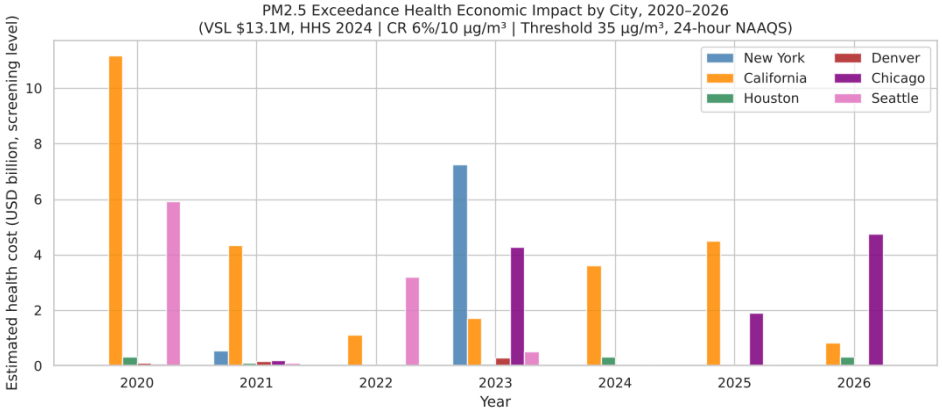



**Figure 6.** Estimated health cost of PM2.5 exceedances by city and year (screening level)

Seattle comes back at the top, this time with the highest cost per resident despite its clean average: its extreme smoke years concentrate a large exposure on a small population. The June 2023 Quebec wildfire smoke, a cross-border event with no local emission source, is estimated at USD 5.5 billion in New York and USD 2.6 billion in Chicago when only the event days are counted (6 to 8 June and 27 and 28 June). Counting every day above the limit in 2023 gives USD 7.2 billion and USD 4.3 billion, but that total also includes days outside the event.

**HOW TO READ THESE NUMBERS**

The estimate applies a long-term exposure relationship to short exceedance episodes and uses a single value of a statistical life. It is useful to compare cities and to show scale. It is not suitable for regulatory analysis, and it is an order of magnitude, not a forecast.

For employers and occupational health planners, the lesson is that exposure is very uneven between years: in the Los Angeles area the monitors recorded PM2.5 above the limit on 15 days per year on average, with a peak of 38 in 2020. For insurers, transported smoke is a cross-regional tail risk. For investors and risk analysts, the Los Angeles estimate (USD 27.3 billion, about USD 3.9 billion per year) is an order-of-magnitude benchmark to screen exposure by location.

## Smaller findings that surprised me

**Houston and New Year's Day.** Houston's list of worst days includes 1 January 2020 (AQI 190) and 1 January 2024 (AQI 184), both defined by fine particles. It is most likely fireworks, not the industrial and ozone mechanism I expected for the city. That is an interpretation: the data shows the spike, not its cause.

**A label I had to correct.** I first described Denver's problem as winter thermal inversions. The data said otherwise: summer is consistently its worst season and winter its cleanest, with one smooth ozone peak and no secondary structure. I changed the label to summer photochemical ozone, so the analysis follows the data and not my assumption.

**No petrochemical hotspot.** I expected Houston's industrial mechanism to show up as one dominant county. It does not: Harris (mean PM2.5 of 10.2 micrograms per cubic metre) and Montgomery (10.0) are practically tied at the top, with Galveston and Brazoria (7.9 and 7.8) behind.

**Fine and coarse particles do not move together everywhere.** The PM2.5 and PM10 correlation goes from 0.85 in New York to 0.48 in Houston, with Seattle at 0.72, Chicago at 0.70, California at 0.63 and Denver at 0.52. Houston's weak link fits a more varied mix of sources. New York's strong one rests on far fewer paired days (512 days with both pollutants, against 2,400 in Chicago), so I treat it with caution.

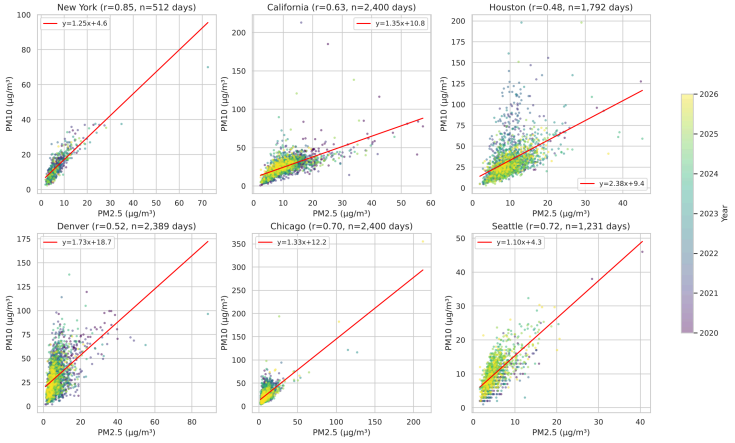



**Figure 7.** Daily PM2.5 against PM10 by city

**An unexplained dip.** Houston's ozone curve has two peaks with a dip in July. I checked whether July simply had fewer observations, and it does not (4,116 against a monthly average of about 3,970). So it looks like a real feature that I could not explain with this data.

## What I could not show

**Known limits.** The comparison covers six cities and six complete years, plus a partial 2026: the daily AQI series ends between 28 February and 1 June 2026 depending on the city, and the pollutant readings run to late July 2026 (except New York's particulates, which end in spring). The 2026 data are preliminary and not yet validated by the EPA. Chicago's smoke episode of 16 to 20 July 2026 falls after the AQI series ends, so it is in the cost figures but not in the AQI trends. "California" is the Los Angeles metropolitan area (two counties), not the state. Population uses the ACS 2022 estimate for every year, summed over the 22 New York counties included here (this may differ slightly from the official Census definition).

## What I would do next

Use yearly population estimates instead of a single year, and test how much the cost estimate changes under other concentration-response and value-of-life assumptions.

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